

Shivang M Raval

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PROFESSIONAL SUMMARY

M.S. Computer Science candidate at Northeastern University with experience in ML infrastructure, multi-agent AI systems, and distributed training. Google Cloud Certified Machine Learning Engineer with a publication in ML deployment strategy.

EDUCATION

Northeastern University, Khoury College of Computer Sciences

M.S. Computer Science. GPA: 3.6/4.0.

Boston, MA

Expected June 2027

Pandit Deendayal Energy University

B.Tech. Computer Science. GPA: 3.5/4.0.

Gandhinagar, India

2021 – 2025

EXPERIENCE

Research Participant, Multi-Agent Systems & Real-Time AI

Google Developer Group (GDG) Cloud Boston

BuildWithAI Workshop

- Built production-grade multi-agent architectures on GCP using Google's ADK and A2A protocol, with quality control via Vertex AI Evaluation Service and Model Armor.
- Engineered a bi-directional streaming agent using the Gemini Live API, Cloud Spanner Graph RAG, and Vertex AI Memory Bank.

AI Engineering Intern, promoted to AI Department Manager

Jan 2025 – June 2025

Webeart AI

- Shipped ML inference systems serving 100K+ daily requests at sub-500ms latency; conducted studies on model generalization and RLHF reward modeling.
- Led a 20-engineer research team, diagnosed bottlenecks in large-scale ML pipelines, and drove changes that raised research throughput by 50%.

Data Science Intern

June 2024 – July 2024

Tor.ai

- Applied regression, ensemble methods, and cross-validation to a forecasting problem, achieving an MAE of 0.12 (20% better than baseline).

PROJECTS

Distributed ML Training Platform

- Built distributed training infrastructure achieving a 10.6x speedup (8h to 45min) via PyTorch DDP; containerized with Docker/Kubernetes and automated CI/CD, cutting model refresh from 4 days to 3 hours across 8 workflows.

Production RAG Infrastructure

- Built a low-latency RAG pipeline on Vertex AI and Gemini with a FastAPI serving layer, Redis caching (60% hit rate, sub-10ms cached latency), and FAISS vector indexing.

Limit Order Book & Matching Engine

- Built a price-time priority matching engine in OCaml supporting limit, market, and cancel orders with IOC/FOK/GTC time-in-force and O(1) cancellation via order-id hash index.
- Verified engine invariants via property-based testing (QCheck) over thousands of randomized sequences; benchmarked at 2M+ ops/sec over TCP on Eio.

AI Infrastructure Stack: Entity Resolution, RL Gym & Policy Classifier

- Fine-tuned a multilingual encoder (InfoNCE loss) for cross-lingual entity resolution across three languages, reaching P@1 = 1.0 on held-out data; shared encoder as reward oracle for a PPO agent and backbone for a zero-shot Gemini-based policy classifier.

CERTIFICATIONS

- Google Cloud Certified: Machine Learning Engineer

SKILLS

Languages: Python, C++, Java, TypeScript, OCaml, SQL, Bash

ML & Data: PyTorch, RLHF, FAISS, LangChain, Redis, PostgreSQL

Cloud & Infrastructure: GCP (Vertex AI, ADK, Gemini API), AWS, Docker, Kubernetes, SageMaker, CI/CD

Systems: Distributed systems, Concurrency, Type systems, Property-based testing

PUBLICATIONS

- IJISRT 2024: ML Deployment Platforms in Business: Scalable Infrastructure, Operational Challenges, and Deployment Strategies

ORGANIZATIONS

- GDG Cloud Boston, Member, BuildWithAI 2026 Participant